Quantum Cryptography without Entanglements

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“Standard” Quantum Crypto Primitives

Pseudorandom state generators (PRSG) $k \mapsto |\phi_k\rangle$

$$|\phi_k\rangle^{\otimes poly} \approx_c |\psi\rangle^{\otimes poly}$$

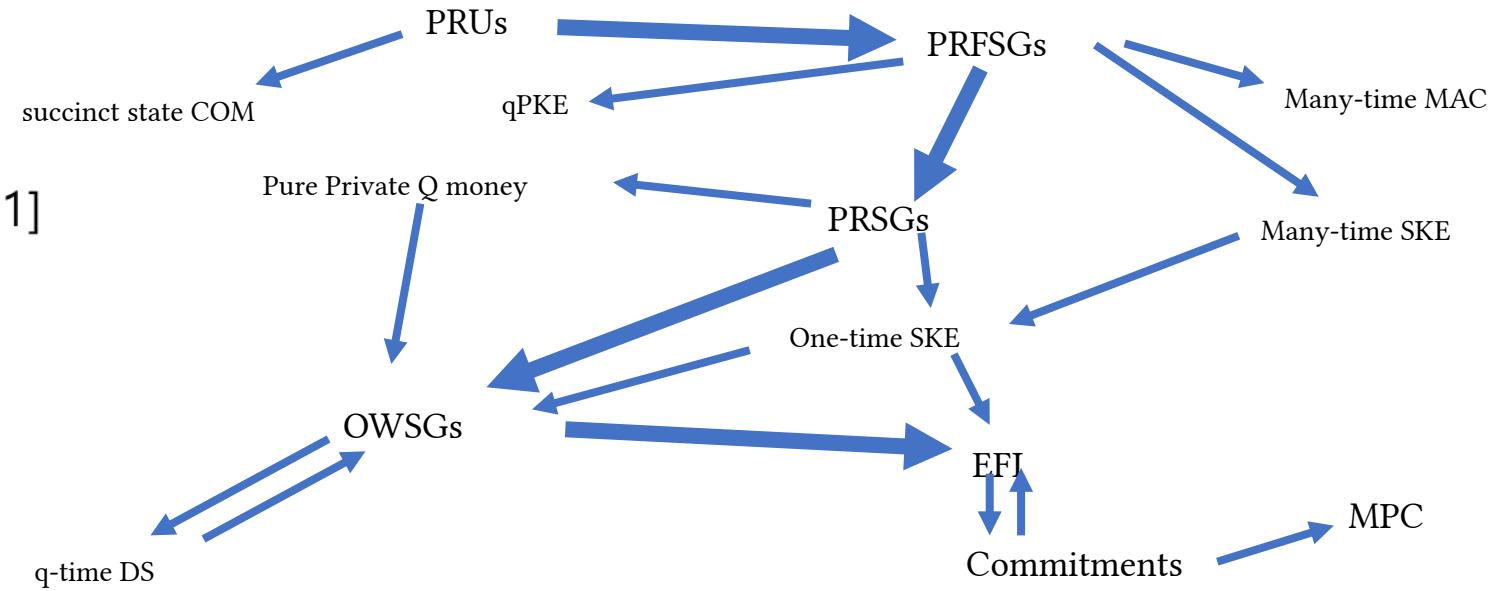
Pseudorandom unitaries (PRU) $k \mapsto U_k$

$$A^{U_k} \approx_c A^{V}$$

Haar random

Can exists even if P=NP [Kre21]

* JLS18, MH25



Disadvantages?

Haar random is

- too difficult to me (mathematically)
- too heavy (doubly exponential)
- too entangled (n-qubits work together)

Few explicit bad things:

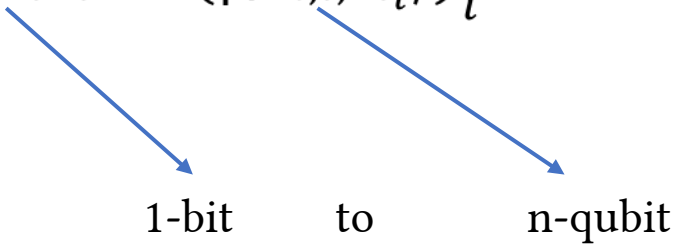
- Implementation requires a stable quantum memory
- Size of Enc/MAC etc from PR(F)S is large (m-bit msg -> mn-qubit ctxt)
- PR(F)S does not imply PRU in a black-box way

States are not operational

Crypto 101: Message **m** and (pseudo)random **r**
then $\boldsymbol{m} \oplus \boldsymbol{r}$ looks like uniform random

With pseudorandom states $|\boldsymbol{\phi_k}\rangle$ and message **m**...??

- Parse many (shorter) PRSs $|\phi_{k,i,b}\rangle$ and $(m_i)_i \mapsto \left(\left|\phi_{k,i,m_i}\right\rangle\right)_i$



PRU is better here but difficult, heavy, entangled

This work: New definitions
Product of 1-qubit random unitaries

Product pseudorandom unitaries (⊗PRU)

$$k \mapsto U_1^{(k)} \otimes U_2^{(k)} \otimes \cdots \otimes U_n^{(k)}$$
$$= \bigotimes_{i=1}^{n} U_i^{(k)} =: U^{(k)}$$

which is computationally indistinguishable from product 1-qubit Haar unitaries

- Operational!

$$U^{(k)}|m_1 \ldots m_n\rangle = \bigotimes_{i=1}^{n} U_i^{(k)}|m_i\rangle$$

- No entanglement

* The name ⊗PRU is suggested by Nehoran Barak

Pseudo OTP from ⊗PRU

Product pseudorandom unitaries (⊗**PRU**)

$$k \mapsto \bigotimes_{i=1}^{n} U_i^{(k)} =: U^{(k)}$$

which is computationally indistinguishable from product 1-qubit unitaries

Pseudo OTP (n-bit msg to n-qubit ctxt)

$$(k, m) \mapsto U^{(k)}|m\rangle = \bigotimes_{i=1}^{n} U_i^{(k)}|m_i\rangle$$
$$\approx_c \bigotimes_{i=1}^{n} |\psi_i\rangle$$



"classical mode"

$$\mapsto \left(\text{description}(\psi_i)\right)_i$$

by efficient tomography

⊗PRU Function

iid 1-qubit Haar

Product pseudorandom unitary functions (⊗**PRUF**)

$$(k, x) \mapsto \bigotimes_{i=1}^{n} U_i^{(k,x)} =: U^{(k,x)}$$

Security: This map is computationally indistinguishable "ideal" map $x \mapsto \bigotimes_{i=1}^{n} V_i^{(x)}$

Note: Output length is not very important

This gives quantum analogous of "textbook" primitives from PRF

- IND-CPA symmetric encryption $m \mapsto (r, U^{(k,r)}|m\rangle)$
- EUF-CMA MAC $m \mapsto U^{(k,m)}|0\rangle$

Textbook-level quantum crypto, with classical mode!

Quantum-accessible ⊗PRUF ⇒ PRSG

Quantum-accessible ⊗PRUF: Given key k,

$$\sum_x \alpha_x |x, \phi\rangle \mapsto \sum_x \alpha_x |x\rangle \otimes U^{(k,x)}|\phi\rangle$$

1-qubit

Tricky identity: for $|\Psi^-\rangle = \frac{|01\rangle - |10\rangle}{\sqrt{2}}$,

$$(U \otimes U)|\Psi^-\rangle = (\det U)|\Psi^-\rangle$$

- ⊗PRUF gives pseudorandom diagonal unitary $\sum_x e^{\phi_{x,k}}|x\rangle\langle x| = D_k$

$$|x\rangle \otimes |\Psi^-\rangle \mapsto |x\rangle \otimes \left(U^{(k,x)} \otimes U^{(k,x)}\right)|\Psi^-\rangle = (\det U^{(k,x)}\,|x\rangle) \otimes |\Psi^-\rangle$$

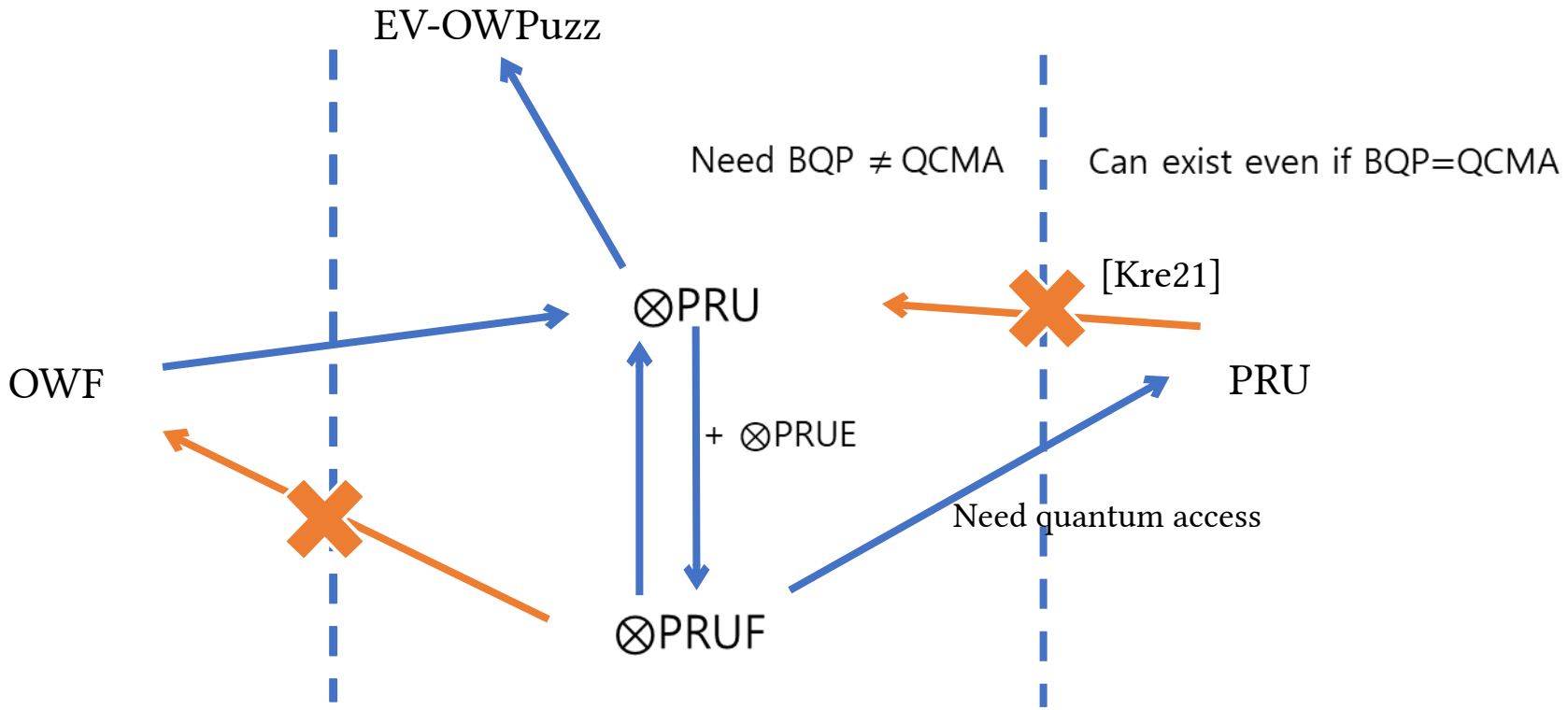
- Pseudorandom diagonal gives PRSG: $D_k|+\rangle^{\otimes n}$

Similarly ⊗PRUF give PRFSG and PRU as well [AGQY22, GKT26]

Not written in the paper… DHDHDHD

* Coherent access is necessary [HOV26]

Landscapes & Further results



⊗PRUE maps few 1-qubit unitaries to many 1-qubit unitaries (eg PRG)

Candidate constructions: Given public 1-qubit unitaries $P_1, \ldots, P_m$, construction uses e.g.

$$P_i \circ P_j \circ P_k, \qquad P_{1,x_1} \circ P_{2,x_2} \circ P_{3,x_3} \circ \cdots$$

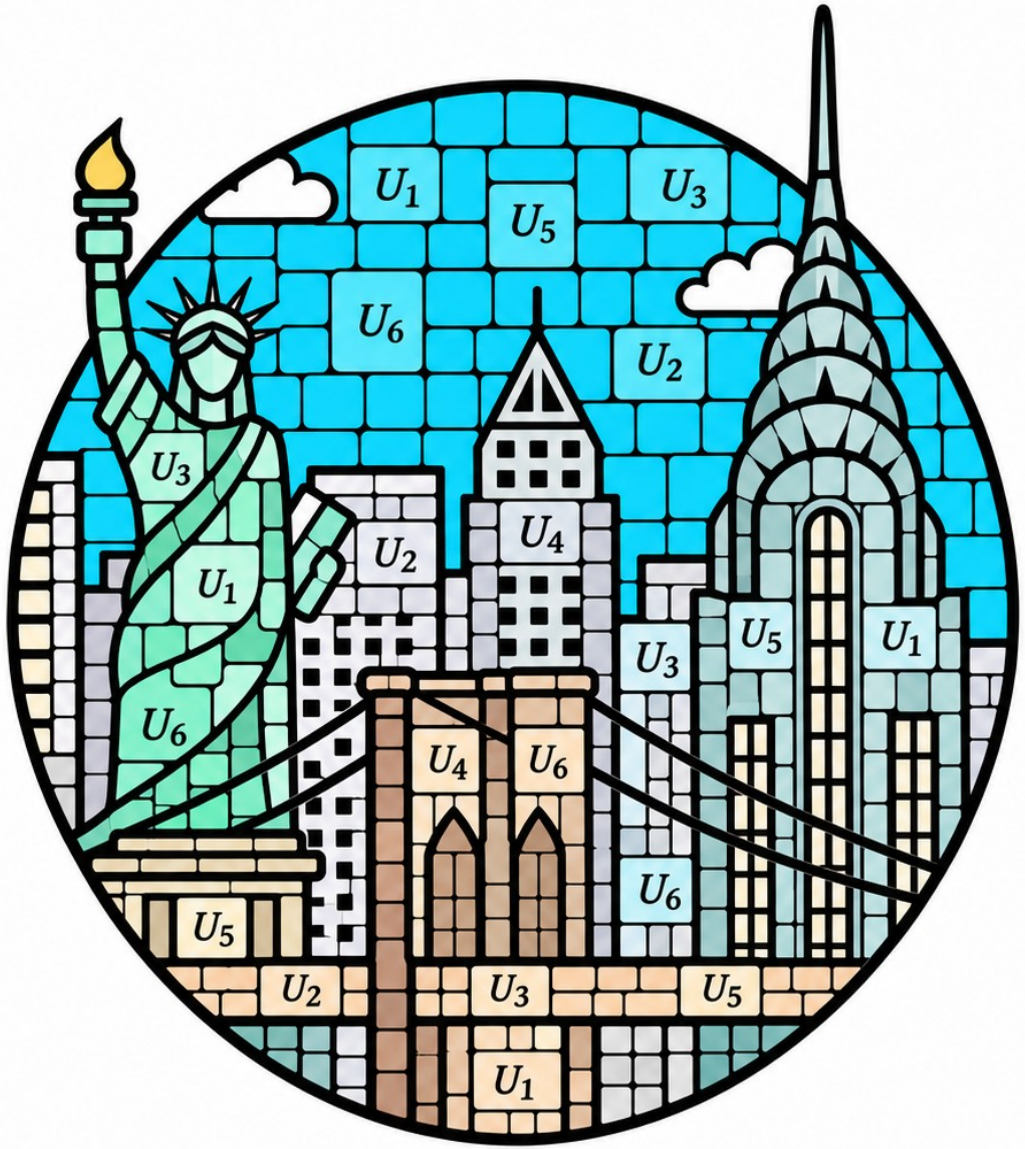
Future directions

(some solved by AI already…)

- Further applications/powers of ⊗PRU and friends?
- Implementation in real-world?
- Application of $(U \otimes U)|\Psi^-\rangle = (\det U)|\Psi^-\rangle$?
- Power of pseudorandom diagonal unitaries?

Thanks!

Any question?



Appendix: Separation

[Theorem] (Informal) Let F be ⊗PRUF. For any efficient binary output algorithm A^F, either

- A^F is not "almost" deterministic
- A^F is not secure pseudorandom generator

Idea: Space of F can be seen as a product of single-qubit unitaries

$$f: (product) \to [0,1] \text{ where } f(F) = \Pr[A^F \to 1]$$

- If f is highly concentrated on 0 or 1 whp, then we can use fake G' (ie G is useless)
- If f occasionally lies "middle", it means A is not deterministic
- If f is concentrated on 0 and 1 with "small middle"?
  Impossible because a new isoperimetric inequality!